# GLD/GDX Pairs Trading

## Cointegration Analysis & Kalman Filter Spread

*Period: 2010-2024 | Source: Yahoo Finance via yfinance*

## 1. Dataset & Motivation

GLD (SPDR Gold Trust) and GDX (VanEck Gold Miners ETF) are two of the most liquid gold-linked instruments available. GLD tracks spot gold directly; GDX holds gold mining companies and is driven by gold prices but also carries equity-specific risk -- operating costs, capital structure, and sector sentiment all affect GDX independently of where gold trades. Because both share the same underlying commodity, their prices should move together over long horizons. If that relationship is stable and mean-reverting, deviations from it can be traded: buy the underperformer, short the outperformer, and wait for the gap to close. The goal of this project is to test whether that holds over 2010-2024, understand when and why it breaks down, and design a signal that only fires when the statistical evidence supports it.

## 2. Methods Overview

- **Data collection & EDA:**
  Daily adjusted close from Yahoo Finance. Normalized price charts, rolling correlation, volatility, and return distributions built intuition before any formal testing.
- **Stationarity testing (ADF):**
  Augmented Dickey-Fuller tests confirmed both log price series are I(1) -- non-stationary in levels, stationary in first differences. Required before cointegration testing.
- **Cointegration testing:**
  Engle-Granger and Johansen tests on the full period, then on rolling 2-year windows to map when the relationship was active vs. broken.
- **Kalman filter:**
  Rather than a fixed hedge ratio, the Kalman filter updates its estimate at every timestep -- continuously re-learning the relationship rather than assuming it is fixed.
- **Signal generation:**
  Z-score thresholds on the Kalman spread, gated by the rolling cointegration test so signals only fire when the statistical relationship is confirmed active.

## 3. Key Findings

The headline result is that GLD and GDX are not cointegrated over the full 2010-2024 period (Engle-Granger p = 0.45, Johansen trace = 10.59 vs. 15.49 critical). This is not a failure -- it is the most interesting finding. GLD approximately doubled over 14 years while GDX declined ~40% from its 2011 peak. That permanent divergence cannot be closed by a mean-reverting spread.

- **Regime-dependent relationship**
  Cointegrated in ~79% of rolling 2-year windows despite the full-period result. Tradeable for extended stretches, but the 2013 gold crash and 2022 rate hike cycle periodically broke it down.
- **Static hedge ratio is unstable**
  Rolling OLS hedge ratio ranged from -0.10 to 1.15 across the sample. A single static estimate of 0.34 is wrong for most individual periods -- directly motivating a time-varying approach.
- **Kalman filter resolves the non-stationarity**
  Dynamic spread ADF $p < 0.0001$ vs. 0.87 for the static spread. The Kalman beta tracked rolling OLS beta closely but without the 504-day window lag.
- **ARIMA out-of-sample accuracy**
  ARIMA(1,0,1) on the Kalman spread beat a naive random walk by 23.3% on the holdout period (RMSE 0.00144 vs. 0.00188), confirming short-term autocorrelation the z-score signal exploits.
- **Regime-aware signals are more coherent**
  110 trades generated (59 long, 51 short), with 88.7% of days flat or gated. Signals suppressed during non-cointegrated windows where a naive z-score would trade into a trendng spread.

## 4. Limitations & Reflection

Key caveats on what this analysis does and doesn't show:

- **Look-ahead bias in initialisation**
  The Kalman filter is warm-started with the full-period OLS beta -- in live trading a burn-in period would be needed, introducing mild look-ahead bias in the earliest sample.
- **No transaction costs**
  Bid-ask spread, market impact, and borrowing costs for the short leg are not modelled. Short ETF costs are non-trivial for multi-month holds.
- **Thresholds not optimised**
  +/-2 sigma entry and +/-0.5 sigma close are standard but not calibrated to this pair. In-sample tuning risks overfitting given the limited number of regime windows.
- **Recent non-cointegration**
  2022-2024 windows show weak or absent cointegration. Under the regime gate the strategy would currently be flat -- the statistical foundation for a live trade is not present.
- **No PnL or Sharpe ratio**
  This project evaluates the statistical foundation for a pairs trade, not profitability. Return attribution, sizing, and risk-adjusted performance are outside scope.

## 5. Key Figures

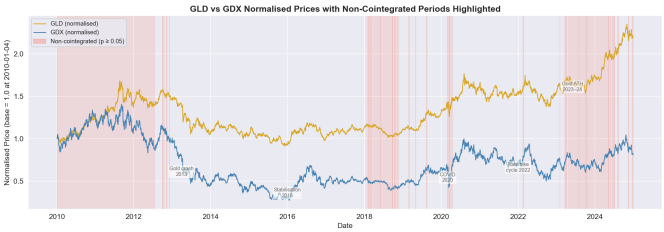


*Fig 1: Normalised prices. Red = non-cointegrated windows. Divergence post-2013 drives the full-period result.*

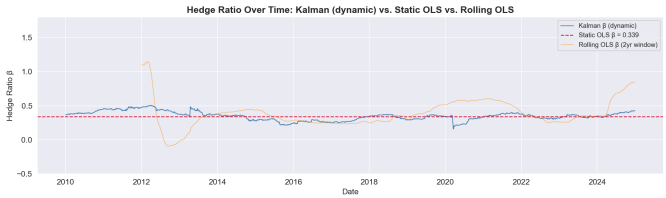


*Fig 2: Hedge ratio over time. Red = static OLS (0.34). Blue = Kalman dynamic (0.16-0.51). Orange = rolling OLS.*

*Fig 3: Kalman z-score with regime gating. Grey = non-cointegrated (signals suppressed). Entry +/-2s, close +/-0.5s.*

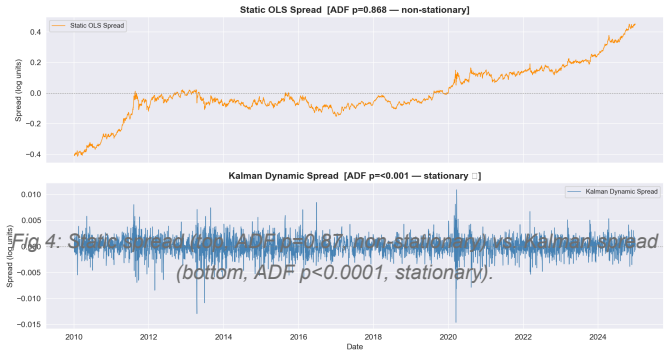


*Fig 4: Static spread (top, ADF p=0.87, non-stationary) vs Kalman spread (bottom, ADF p<0.0001, stationary).*